

TMG (Betaine Anhydrous)

What TMG Is

Trimethylglycine (TMG), sold as betaine anhydrous, is a naturally occurring compound found in beets, spinach, and whole grains. It functions as a methyl donor in the liver, converting homocysteine back into methionine via the enzyme betaine-homocysteine methyltransferase (BHMT). A separate line of research shows it also affects muscle cell hydration and body composition during resistance training. These are two distinct use cases with two distinct dosing protocols.

Protocol 1 – Methylation & Homocysteine Support

Typical studied dose: 1.5g to 3g of betaine anhydrous per day.

Timing: split into two doses, taken with meals (e.g. 1g with breakfast, 1g with dinner) to reduce any digestive discomfort.

Duration: effects on fasting homocysteine are typically measurable within 4-6 weeks of consistent use.

Note: some trials found modest increases in LDL cholesterol and triglycerides at these doses. If you supplement long-term for homocysteine, periodic lipid panel checks are a reasonable precaution.

Protocol 2 – Exercise Performance & Body Composition

Studied dose: 2.5g of betaine anhydrous per day, taken consistently regardless of training day.

Timing: with or without food; pre-workout timing has not shown superiority over any-time daily dosing in the available trials.

Duration: studies showing lean mass and work capacity improvements ran 6 to 9 weeks.

Evaluate results at the 6-week mark, not sooner.

Buying & Quality Checklist

Label Checklist

- Confirm the label reads "betaine anhydrous" or "trimethylglycine (TMG)" — not "betaine HCl," which is an unrelated digestive-acid product.
- Prefer products with third-party testing certification (NSF, Informed Sport, or USP verified) when available.
- Check elemental dose per serving against the protocol you're following — many capsules are 500-750mg, requiring 2-4 capsules to reach 1.5-3g.
- Avoid proprietary blends that combine TMG with undisclosed amounts of other ingredients — you can't dose accurately if you don't know the actual TMG content.
- Vegetarian capsule shells and non-GMO sourcing are common quality signals but not substitutes for third-party testing.

Quick Reference Table

Goal	Daily Dose	Timing	Evaluation Window
Homocysteine / methylation	1.5g - 3g	Split, with meals	4-6 weeks
Exercise performance	2.5g	Any time, daily	6-9 weeks

A Note on Combining With Folate/B12

Betaine's remethylation pathway (via BHMT) runs independently of the folate and B12-dependent methylation cycle. This means TMG can be a useful addition alongside — not a replacement for — adequate folate and B12 status, particularly for anyone with an MTHFR variant or diagnosed elevated homocysteine.